OVER A RING WITH FINITE-DIMENSIONAL NOETHERIAN MAXIMUM SPECTRUM A FINITELY PRESENTED MODULE WITH A LARGE FREE SUMMAND LOCALLY SPLITS OFF A FREE SUMMAND GLOBALLY

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Editorial note. The counted statement span includes the author's own \begin{reference} block citing the published theorem; it is delivered verbatim and produces no unresolved citation. The author writes the enumeration of the specialization of the splitting proposition in full ('with F = empty, h = 0 and no s_i, n = 0 and no x_i, v_i, and k = d + 1'); that sentence is reproduced unchanged. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Spreading out a local direct summand to a principal open, via the colimit descriptions of localization and of finitely presented modules. used at proof lines 38933-38938. Converts the local rank hypothesis into the global dimension statement about V(x).
- The splitting proposition producing an element whose nonvanishing locus is controlled by a codimension bound. invoked at proof lines 38941-38945. Is the main engine of the proof, applied with k = d + 1.
- Universal injectivity checked at stalks, and universally injective maps are split injective. invoked at proof lines 38947-38951. Upgrades the pointwise summand statement into a genuine direct sum decomposition of M.

Source reference. Original lemma 05N7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7296–7296. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00CR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 471–471. Cross-chapter navigation pointer into the

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Source reference. Original lemma 058L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6784–6784. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05CL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6924–6924. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 38731: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38731–38731. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original proposition 38809: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38809–38809. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Theorem 0.2. *Let R be a ring whose max spectrum $\Omega \subset \mathrm{Spec}(R)$ is a Noetherian topological space of dimension $d < \infty$. Let M be a finitely presented R-module such that for all $\mathfrak{m} \in \Omega$ the $R_\mathfrak{m}$-module $M_\mathfrak{m}$ has a free direct summand of rank $> d$. Then $M \cong R \oplus M'$.*

Proof. For $\mathfrak{m} \in \Omega$ suppose that $R_\mathfrak{m}^{\oplus r}$ is a direct summand of $M_\mathfrak{m}$. Then by Algebra, Lemmas 00CR and 05N7 we see that $R_f^{\oplus r}$ is a direct summand of M_f for some $f \in R$, $f \notin \mathfrak{m}$. Hence the assumption means that $\dim V(x) > d$ for all $x \in \Omega$ where $V(x)$ is as in Lemma 38731. By Proposition 38809 applied with $F = \emptyset$, $h = 0$ and no s_i, $n = 0$ and no x_i, v_i, and $k = d + 1$ we find an $s \in M$ and $F' \subset \Omega$ such that every irreducible component of F' has codimension $\geq d + 1$ and $Z(s) \subset F'$. Since $d = \dim(\Omega)$ this forces $F' = \emptyset$. Hence $s : R \to M$ is the inclusion of a direct summand at all maximal ideals. It follows that s is universally injective, see Algebra, Lemma 05CL. Then s is split injective by Algebra, Lemma 058L. $\square$